数据库系统实验报告

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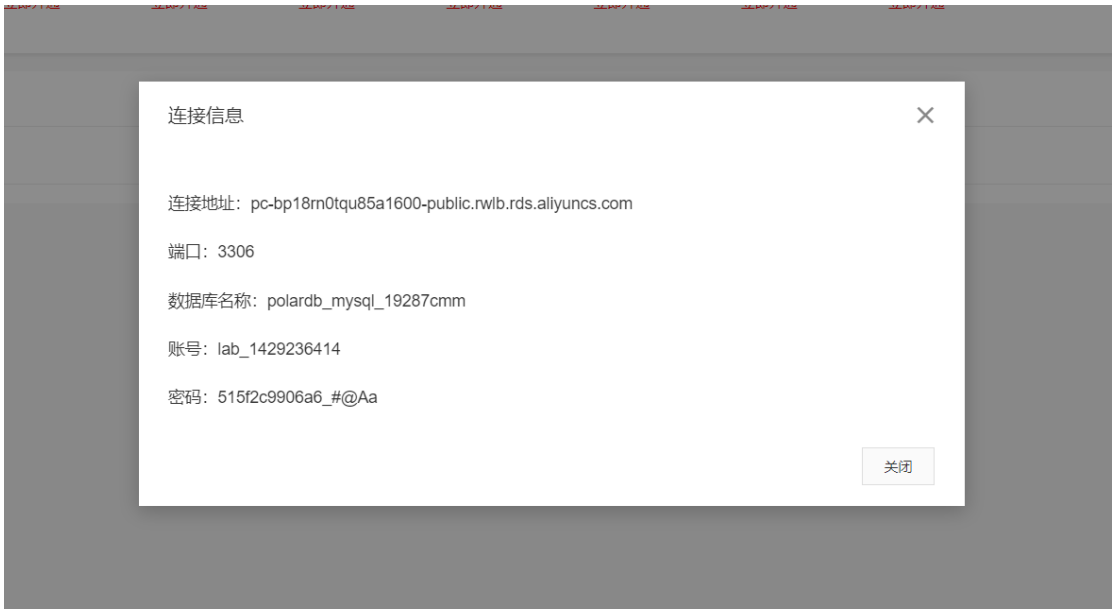
线上实验：阿里 PolarDB 与 XDSQL

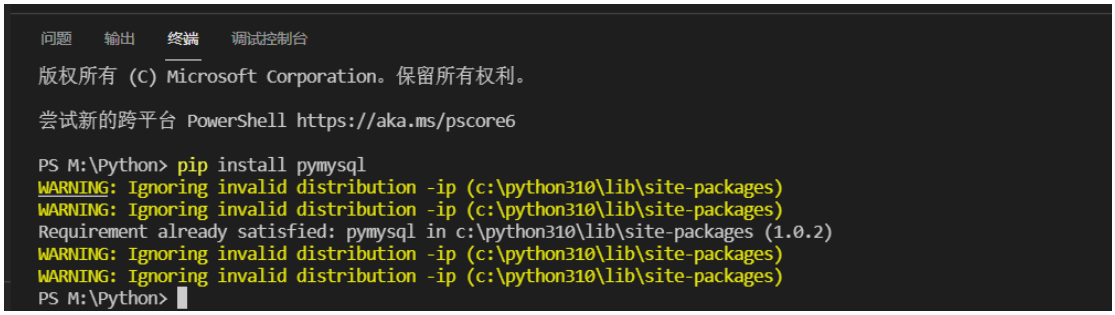
1. PolarDB 远程连接

按照实验步骤要求申请相应的数据库连接信息



用 Python 语言实现对数据库的远程访问

- 安装相应的环境



- 完成配置文件 connect.py

```
DataBase_exp > connect.py > ...
#coding:utf-8

import py
connection = pymysql.connect(host='pc-bp18rn0tqu85a1600-public.rwlb.rds.aliyuncs.com',
                             port=3306,
                             user='lab_1429236414',
                             passwd='515f2c9906a6_#@Aa',
                             db='polardb_mysql_19287cmm')

connectio

try:
(variable) connection: connect
try:
    with connection.cursor() as cursor:
        # 以查询courses表为例
        sql = "SELECT * FROM `courses`"
        cursor.execute(sql)
        for result in cursor:
            # 输出查询结果
            print(result)
finally:
    connection.close()
```

- python connect.py 执行 connect 文件远程连接数据库，内容显示如下。

```
PS M:\Python\DataBase_exp> python .\connect.py
(10, 'Chinese')
(20, 'Mathematics')
(30, 'Physics')
(40, 'English')
(50, 'Chemistry')
(60, 'Biology')
(70, 'Politics')
PS M:\Python\DataBase_exp>
```

2. XDSQL 与 PolarDB 对比实验

1.1 XDSQL 执行的 SQL 语句及结果。

1. 创建数据库

- 创建数据库：

  CREATE DATABASE xjgl;

```
C:\Users\123\Downloads\Compressed\新
init SQL ok
SQL>>CREATE DATABASE xjgl;
create database ok!
SQL>>
```

2. 切换数据库

- 切换数据库：

USE DATABASE xjgl;

```
SQL>>USE DATABASE xjgl;
use xjgl
SQL>>
```

3. 定义模式 / 删除模式

- 定义基本表：

创建学生表 Student、课程表 Course、学生选课表 SC，SQL 语句：

CREATE TABLE Student(Sno CHAR(9),Sname CHAR(20),Ssex CHAR(2),Sage INT , Sdept CHAR(20));

CREATE TABLE Course(Cno CHAR(4),Cname CHAR(40),Cpno CHAR(4) , Ccredit INT);

CREATE TABLE SC(Sno CHAR(9),Cno CHAR(4),Grade INT);

```
SQL>>CREATE TABLE Student(Sno CHAR(9),Sname CHAR(20),Ssex CHAR(2),Sage INT , Sdept CHAR(20));
create table ok
SQL>>

SQL>>CREATE TABLE Course(Cno CHAR(4),Cname CHAR(40),Cpno CHAR(4) , Ccredit INT);
create table ok
SQL>>CREATE TABLE SC(Sno CHAR(9),Cno CHAR(4),Grade INT);
create table ok
SQL>>
```

- 删除 Student 表：

DROP TABLE Student;

```
SQL>>DROP TABLE Student;
drop table ok
SQL>>
```

- 修改基本表：

向 Student 表增加”出生日期“列，数据类型为日期型：

ALTER TABLE Student ADD birthdate DATE;

向 Student 表增加 Sno 列为主码的约束条件：

ALTER TABLE Student ADD PRIMARY KEY(Sno);

向 Student 表增加 Sname 列为唯一值的约束条件：

ALTER TABLE Student ADD UNIQUE(Sname);

```
SQL>>ALTER TABLE Student ADD birthdate DATE;
语法错误/不支持本条SQL语句
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SQL>>ALTER TABLE Student ADD PRIMARY KEY(Sno);
语法错误/不支持本条SQL语句
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SQL>>ALTER TABLE Student ADD UNIQUE(Sname);
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```

4. 完整性约束条件

- 重新创建带有完整性约束条件的用户数据表：

CREATE TABLE Student(Sno CHAR(9) PRIMARY KEY,Sname CHAR(20) UNIQUE , Ssex CHAR(2),Sage SMALLINT,Sdept CHAR(20));

CREATE TABLE Course(Cno CHAR(4) PRIMARY KEY,Cname CHAR(40) NOT NULL,Cpno CHAR(4),Ccredit INT,FOREIGN KEY(Cpno) REFERENCES Course(Cno));

CREATE TABLE SC(Sno CHAR(9),Cno CHAR(4),Grade INT,PRIMARY KEY(Sno,Cno),FOREIGN KEY(Sno)REFERENCES Student(Sno) , FOREIGN KEY(Cno)REFERENCES Course(Cno));

```
SQL>>CREATE TABLE Student(Sno CHAR(9) PRIMARY KEY,Sname CHAR(20) UNIQUE , Ssex CHAR(2),Sage SMAL
LINT,Sdept CHAR(20));
语法错误/不支持本条SQL语句
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```

5. 索引

- 建立索引

CREATE UNIQUE INDEX Stusno ON Student(Sno);

CREATE UNIQUE INDEX Coucno ON Course(Cno);

CREATE UNIQUE INDEX SCno ON SC(Sno ASC,Cno DESC);

```
SQL>>CREATE UNIQUE INDEX Stusno ON Student(Sno);
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语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句

SQL>>CREATE UNIQUE INDEX Coucno ON Course(Cno);
语法错误/不支持本条SQL语句
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语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句

SQL>>CREATE UNIQUE INDEX SCno ON SC(Sno ASC,Cno DESC);
语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句
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语法错误/不支持本条SQL语句
```

- 删除索引

DROP INDEX Stusno ON Student;

```
SQL>>DROP INDEX Stusno ON Student;
语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句
```

6. 数据查询

- 插入数据

INSERT INTO Student VALUES('201215121','李勇','男',20,'CS');

INSERT INTO Student VALUES('201215122','刘晨','女',19,'CS');

INSERT INTO Student VALUES('201215123','王敏','女',18,'MA');

INSERT INTO Student VALUES('201215124','张立','男',19,'IS');

INSERT INTO COURSE VALUES('1','数据库','5',4);

INSERT INTO COURSE VALUES('3','信息系统','1',4);

INSERT INTO COURSE VALUES('4','操作系统','6',3);

INSERT INTO COURSE VALUES('5','数据结构','7',4);

INSERT INTO COURSE VALUES('7','PASCAL 语言','6',4);

INSERT INTO SC VALUES('201215121','1',92);

INSERT INTO SC VALUES('201215121','2',85);

INSERT INTO SC VALUES('201215121','3',88);

INSERT INTO SC VALUES('201215122','2',90);

INSERT INTO SC VALUES('201215122','3',80);

删除了 Student 表

```
SQL>>INSERT INTO Student VALUES('201215121','李勇','男',20,'CS');
student is not Exist,Drop Table error
SQL>>INSERT INTO Student VALUES('201215122','刘晨','女',19,'CS');
student is not Exist,Drop Table error
SQL>>INSERT INTO Student VALUES('201215123','王敏','女',18,'MA');
student is not Exist,Drop Table error
SQL>>INSERT INTO Student VALUES('201215124','张立','男',19,'IS');
student is not Exist,Drop Table error
SQL>>

SQL>>INSERT INTO COURSE VALUES('1','数据库','5',4);
insert ok
SQL>>INSERT INTO COURSE VALUES('3','信息系统','1',4);
insert ok
SQL>>INSERT INTO COURSE VALUES('4','操作系统','6',3);
insert ok
SQL>>INSERT INTO COURSE VALUES('5','数据结构','7',4);
insert ok
SQL>>INSERT INTO COURSE VALUES('7','PASCAL语言','6',4);
insert ok
SQL>>

SQL>>INSERT INTO SC VALUES('201215121','1',92);
insert ok
SQL>>INSERT INTO SC VALUES('201215121','2',85);
insert ok
SQL>>INSERT INTO SC VALUES('201215121','3',88);
insert ok
SQL>>INSERT INTO SC VALUES('201215122','2',90);
insert ok
SQL>>INSERT INTO SC VALUES('201215122','3',80);
insert ok
SQL>>
```

重新添加了 Student 表

```
SQL>>CREATE TABLE Student(Sno CHAR(9),Sname CHAR(20),Ssex CHAR(2),Sage INT , Sdept CHAR(20));
create table ok
SQL>>INSERT INTO Student VALUES('201215121','李勇','男',20,'CS');
insert ok
SQL>>INSERT INTO Student VALUES('201215122','刘晨','女',19,'CS');
insert ok
SQL>>INSERT INTO Student VALUES('201215123','王敏','女',18,'MA');
insert ok
SQL>>INSERT INTO Student VALUES('201215124','张立','男',19,'IS');
insert ok
SQL>>
```

INSERT INTO COURSE VALUES('2','数学',NULL,2);

INSERT INTO COURSE VALUES('6','数据处理',NULL,2);

```
SQL>>INSERT INTO COURSE VALUES('2','数学',NULL,2);
语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句

SQL>>INSERT INTO COURSE VALUES('6','数据处理',NULL,2);
语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句
```

insert into course(Cno,Cname,Ccredit) values ('2','数学',2);

insert into course(Cno,Cname,Ccredit) values ('6','数据处理',2);

```
SQL>>insert into course(Cno,Cname,Ccredit) values ('2','数学',2);
insert ok
SQL>>insert into course(Cno,Cname,Ccredit) values ('6','数据处理',2);
insert ok
SQL>>
```

- 更新数据

UPDATE Student SET Sage=Sage+1;

```
SQL>>UPDATE Student SET Sage=Sage+1;
语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句
```

UPDATE Course SET Cpno='1' WHERE Cno='6';

```
SQL>>UPDATE Course SET Cpno='1' WHERE Cno='6';
update 1 rows
SQL>>
```

- 删除数据

DELETE FROM SC WHERE Sno='201215121' AND Cno='1';

DELETE FROM SC;

```
SQL>>DELETE FROM SC WHERE Sno='201215121' AND Cno='1';
delete 1 rows
SQL>>DELETE FROM SC;
delete 4 rows
SQL>>
```

- 查询数据

单表查询

SELECT * FROM Student;

SELECT * FROM Sc;

SELECT * FROM Course;

SELECT Sno,Sname,Sage FROM Student;

SELECT Sname FROM Student WHERE Sdept='cs';

SELECT Sname,Sage FROM Student WHERE Sage<20;

```
SQL>>SELECT * FROM Student;
sno       sname    ssex     sage     sdept
201215121          李勇     男       20       CS
201215122          刘晨     女       19       CS
201215123          王敏     女       18       MA
201215124          张立     男       19       IS
Total row: 4
SQL>>SELECT * FROM Sc;
sno       cno      grade
Total row: 0
SQL>>SELECT * FROM Course;
cno       cname    cpno     ccredit
1         数据库   5        4
3         信息系统          1        4
4         操作系统          6        3
5         数据结构          7        4
7         PASCAL语言        6        4
2         数学     NULL     2
6         数据处理          1        2
Total row: 7
SQL>>

SQL>>SELECT Sno,Sname,Sage FROM Student;
sno       sname    sage
201215121          李勇     20
201215122          刘晨     19
201215123          王敏     18
201215124          张立     19
Total row: 4
SQL>>SELECT Sname FROM Student WHERE Sdept='cs';
sname
Total row: 0
SQL>>SELECT Sname,Sage FROM Student WHERE Sage<20;
sname    sage
刘晨     19
王敏     18
张立     19
Total row: 3
SQL>>
```

SELECT Sname,2021-Sage FROM Student;

```
SQL>>SELECT Sname,2021-Sage FROM Student;
语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句
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语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句
```

SELECT Sname,'Year of birth:',2014-Sage FROM Student;

```
SQL>>SELECT Sname,'Year of birth:',2014-Sage FROM Student;
语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句
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语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句
```

SELECT DISTINCT Sno FROM SC;

SELECT ALL Sno FROM SC;

```
SQL>>SELECT DISTINCT Sno FROM SC;
语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句

SQL>>SELECT ALL Sno FROM SC;
语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句
```

SELECT* FROM Student WHERE Sage BETWEEN 20 AND 21;

SELECT * FROM Student WHERE Sdept IN('CS','IS');

SELECT * FROM Student WHERE Sname LIKE '刘%';

SELECT * FROM Student WHERE Sname LIKE '刘_';

```
SQL>>SELECT* FROM Student WHERE Sage BETWEEN 20 AND 21;
语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句

SQL>>SELECT * FROM Student WHERE Sdept IN('CS','IS');
IN语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句

SQL>>SELECT * FROM Student WHERE Sname LIKE '刘%';
LIKE语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句

SQL>>SELECT * FROM Student WHERE Sname LIKE '刘_';
LIKE语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句
```

SELECT* FROM Student ORDER BY Sdept,Sage DESC;

SELECT COUNT(*) FROM Student;

SELECT COUNT(DISTINCT Sno) FROM SC;

SELECT AVG(Sage) FROM Student;

SELECT SUM(Ccredit) FROM Course;

SELECT Sno FROM SC GROUP BY SNO HAVING COUNT(*)>=3;

```
SQL>>SELECT* FROM Student ORDER BY Sdept,Sage DESC;
语法错误/不支持本条SQL语句
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语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句

SQL>>SELECT COUNT(*) FROM Student;
语法错误/不支持本条SQL语句
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语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句

SQL>>SELECT COUNT(DISTINCT Sno) FROM SC;
语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句

SQL>>SELECT AVG(Sage) FROM Student;
语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句
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SQL>>SELECT SUM(Ccredit) FROM Course;
语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句
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SQL>>SELECT Sno FROM SC GROUP BY SNO HAVING COUNT(*)>=3;
语法错误/不支持本条SQL语句
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```

多表查询

SELECT Student.Sno,Sname FROM Student,SC WHERE Student.Sno=SC.Sno AND SC.Cno='2';

```
SQL>>SELECT Student.Sno,Sname FROM Student,SC WHERE Student.Sno=SC.Sno
AND SC.Cno='2';
student.sno      sname
Total row: 0
```

SELECT FIRST.Cno,SECOND.Cpno FROM Course FIRST,Course SECOND WHERE FIRST.Cpno=SECOND.Cno;

SELECT* FROM Student LEFT OUTER JOIN SC ON(Student.Sno=SC.Sno);

SELECT* FROM Course RIGHT OUTER JOIN SC ON(Course.Cno=SC.Cno);

SELECT* FROM Course JOIN SC ON(Course.Cno=SC.Cno);

```
SQL>>SELECT FIRST.Cno,SECOND.Cpno FROM Course FIRST,Course SECOND WHERE
 FIRST.Cpno=SECOND.Cno;
语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句
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语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句

SQL>>SELECT* FROM Student LEFT OUTER JOIN SC ON(Student.Sno=SC.Sno);
语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句
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语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句

SQL>>SELECT* FROM Course RIGHT OUTER JOIN SC ON(Course.Cno=SC.Cno);
语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句
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语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句

SQL>>SELECT* FROM Course JOIN SC ON(Course.Cno=SC.Cno);
语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句
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语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句
```

嵌套查询

SELECT Sname FROM Student WHERE EXISTS(SELECT* FROM SC WHERE Sno=Student.Sno AND Cno='1');

```
SQL>>SELECT Sname FROM Student WHERE EXISTS(SELECT* FROM SC WHERE Sno=S
tudent.Sno AND Cno='1');
EXISTS语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句
```

7. 视图

- 定义视图

CREATE VIEW V1 AS SELECT SC.Sno,Sname FROM Student,SC WHERE Sdept='IS' AND SC.Cno='1' AND Student.Sno=SC.Sno;

CREATE VIEW V2 AS SELECT Sname FROM V1;

```
SQL>>SELECT Sname FROM Student WHERE EXISTS(SELECT* FROM SC WHERE Sno=S
tudent.Sno AND Cno='1');
EXISTS语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句

SQL>>CREATE VIEW V1 AS SELECT SC.Sno,Sname FROM Student,SC WHERE Sdept=
'IS' AND SC.Cno='1' AND Student.Sno=SC.Sno;
语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句
sc.sno  sname
Total row: 0
SQL>>CREATE VIEW V2 AS SELECT Sname FROM V1;
语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句
table not exist
```

- 删除视图

DROP VIEW V2;

```
SQL>>DROP VIEW V2;
语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句
```

- 查询视图

SELECT Sname FROM V1 WHERE Sname LIKE '李_';

```
SQL>>SELECT Sname FROM V1 WHERE Sname LIKE '李_';
LIKE语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句
语法错误/不支持本条SQL语句
```

1.2 PolarDB 执行的 SQL 语句及结果。

1. 创建数据库

- 创建数据库：

CREATE DATABASE xjgl;

文件(F) 编辑(E) 格式(O) 查看(V) 帮助(H)

1 Something is wrong! Please look at the terminal immediately!

```
1
(1044, "Access denied for user 'lab_1429236414'@'%' to database 'xjgl'")
```

2. 切换数据库

- 切换数据库：

USE DATABASE xjgl;

2 Something is wrong! Please look at the terminal immediately!

```
2
(1064, "You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right synta
 1")
```

3. 定义模式 / 删除模式

- 定义基本表：

创建学生表 Student、课程表 Course、学生选课表 SC，SQL 语句：

CREATE TABLE Student(Sno CHAR(9),Sname CHAR(20),Ssex CHAR(2),Sage INT , Sdept CHAR(20));

CREATE TABLE Course(Cno CHAR(4),Cname CHAR(40),Cpno CHAR(4) , Ccredit INT);

CREATE TABLE SC(Sno CHAR(9),Cno CHAR(4),Grade INT);

3 Operation Success!
4 Operation Success!
5 Operation Success!

- 删除 Student 表：

DROP TABLE Student;

- 修改基本表：

向 Student 表增加”出生日期“列，数据类型为日期型：

ALTER TABLE Student ADD birthdate DATE;

向 Student 表增加 Sno 列为主码的约束条件：

ALTER TABLE Student ADD PRIMARY KEY(Sno);

向 Student 表增加 Sname 列为唯一值的约束条件：

ALTER TABLE Student ADD UNIQUE(Sname);

6 Operation Success!
7 Operation Success!
8 Operation Success!

4. 完整性约束条件

- 重新创建带有完整性约束条件的用户数据表：

CREATE TABLE Student(Sno CHAR(9) PRIMARY KEY,Sname CHAR(20) UNIQUE , Ssex CHAR(2),Sage SMALLINT,Sdept CHAR(20));

CREATE TABLE Course(Cno CHAR(4) PRIMARY KEY,Cname CHAR(40) NOT NULL,Cpno CHAR(4),Ccredit INT,FOREIGN KEY(Cpno) REFERENCES Course(Cno));

CREATE TABLE SC(Sno CHAR(9),Cno CHAR(4),Grade INT,PRIMARY KEY(Sno,Cno),FOREIGN KEY(Sno)REFERENCES Student(Sno) , FOREIGN KEY(Cno)REFERENCES Course(Cno));

12 Operation Success!
13 Operation Success!
14 Operation Success!

5. 索引

- 建立索引

CREATE UNIQUE INDEX Stusno ON Student(Sno);

CREATE UNIQUE INDEX Coucno ON Course(Cno);

CREATE UNIQUE INDEX SCno ON SC(Sno ASC,Cno DESC);

15 Operation Success!
16 Operation Success!
17 Operation Success!

- 删除索引

DROP INDEX Stusno ON Student;

18 Operation Success!

6. 数据查询

- 插入数据

INSERT INTO Student VALUES('201215121','李勇','男',20,'CS');

INSERT INTO Student VALUES('201215122','刘晨','女',19,'CS');

INSERT INTO Student VALUES('201215123','王敏','女',18,'MA');

INSERT INTO Student VALUES('201215124','张立','男',19,'IS');

INSERT INTO COURSE VALUES('1','数据库','5',4);

INSERT INTO COURSE VALUES('3','信息系统','1',4);

INSERT INTO COURSE VALUES('4','操作系统','6',3);

INSERT INTO COURSE VALUES('5','数据结构','7',4);

INSERT INTO COURSE VALUES('7','PASCAL 语言','6',4);

INSERT INTO SC VALUES('201215121','1',92);

INSERT INTO SC VALUES('201215121','2',85);

INSERT INTO SC VALUES('201215121','3',88);

INSERT INTO SC VALUES('201215122','2',90);

INSERT INTO SC VALUES('201215122','3',80);

删除了 Student 表

```
10 Something is wrong! Please look at the terminal immediately!
11 Something is wrong! Please look at the terminal immediately!
12 Something is wrong! Please look at the terminal immediately!
13 Something is wrong! Please look at the terminal immediately!
14 Operation Success!
15 Operation Success!
16 Operation Success!
17 Operation Success!
18 Operation Success!
19 Operation Success!
20 Operation Success!
21 Operation Success!
22 Operation Success!
23 Operation Success!
```

```
10
(1146, "Table 'polardb_mysql_19287cmm.student' doesn't exist")
11
(1146, "Table 'polardb_mysql_19287cmm.student' doesn't exist")
12
(1146, "Table 'polardb_mysql_19287cmm.student' doesn't exist")
13
(1146, "Table 'polardb_mysql_19287cmm.student' doesn't exist")
14
15
16
17
18
19
20
21
22
23
```

重新添加了 Student 表

19 Operation Success!
20 Operation Success!
21 Operation Success!
22 Operation Success!
23 Operation Success!
24 Operation Success!
25 Operation Success!
26 Operation Success!
27 Operation Success!
28 Operation Success!
29 Operation Success!
30 Operation Success!
31 Operation Success!
32 Operation Success!

INSERT INTO COURSE VALUES('2','数学',NULL,2);

INSERT INTO COURSE VALUES('6','数据处理',NULL,2);

33 Operation Success!
34 Operation Success!

insert into course(Cno,Cname,Ccredit) values ('2','数学',2);

insert into course(Cno,Cname,Ccredit) values ('6','数据处理',2);

35 Operation Success!
36 Operation Success!

- 更新数据

UPDATE Student SET Sage=Sage+1;

37 Operation Success!

UPDATE Course SET Cpno='1' WHERE Cno='6';

38 Operation Success!

- 删除数据

DELETE FROM SC WHERE Sno='201215121' AND Cno='1';

DELETE FROM SC;

39 Operation Success!
40 Operation Success!

- 查询数据

单表查询

SELECT * FROM Student;

SELECT * FROM Sc;

SELECT * FROM Course;

SELECT Sno,Sname,Sage FROM Student;

SELECT Sname FROM Student WHERE Sdept='cs';

SELECT Sname,Sage FROM Student WHERE Sage<20;

```
41 Operation Success!
('201215121', '李勇', '男', 21, 'CS')
('201215122', '刘晨', '女', 20, 'CS')
('201215123', '王敏', '女', 19, 'MA')
('201215124', '张立', '男', 20, 'IS')
42 Operation Success!
43 Operation Success!
('1', '数据库', '5', 4)
('3', '信息系统', '1', 4)
('4', '操作系统', '6', 3)
('5', '数据结构', '7', 4)
('7', 'PASCAL语言', '6', 4)
('2', '数学', None, 2)
('6', '数据处理', '1', 2)
('2', '数学', None, 2)
('6', '数据处理', '1', 2)
44 Operation Success!
('201215121', '李勇', 21)
('201215122', '刘晨', 20)
('201215123', '王敏', 19)
('201215124', '张立', 20)
45 Operation Success!
('李勇',)
('刘晨',)
46 Operation Success!
('王敏', 19)
```

SELECT Sname,2021-Sage FROM Student;

```
47 Operation Success!
('李勇', 2000)
('刘晨', 2001)
('王敏', 2002)
('张立', 2001)
```

SELECT Sname,'Year of birth:',2014-Sage FROM Student;

```
48 Operation Success!
('李勇', 'Year of birth:', 1993)
('刘晨', 'Year of birth:', 1994)
('王敏', 'Year of birth:', 1995)
('张立', 'Year of birth:', 1994)
```

SELECT DISTINCT Sno FROM SC;

SELECT ALL Sno FROM SC;

```
49 Operation Success!
50 Operation Success!
```

SELECT* FROM Student WHERE Sage BETWEEN 20 AND 21;

SELECT * FROM Student WHERE Sdept IN('CS','IS');

SELECT * FROM Student WHERE Sname LIKE '刘%';

SELECT * FROM Student WHERE Sname LIKE '刘_';

```
51 Operation Success!
('201215121', '李勇', '男', 21, 'CS')
('201215122', '刘晨', '女', 20, 'CS')
('201215124', '张立', '男', 20, 'IS')
52 Operation Success!
('201215121', '李勇', '男', 21, 'CS')
('201215122', '刘晨', '女', 20, 'CS')
('201215124', '张立', '男', 20, 'IS')
53 Operation Success!
('201215122', '刘晨', '女', 20, 'CS')
54 Operation Success!
('201215122', '刘晨', '女', 20, 'CS')
```

SELECT* FROM Student ORDER BY Sdept,Sage DESC;

SELECT COUNT(*) FROM Student;

SELECT COUNT(DISTINCT Sno) FROM SC;

SELECT AVG(Sage) FROM Student;

SELECT SUM(Ccredit) FROM Course;

SELECT Sno FROM SC GROUP BY SNO HAVING COUNT(*)>=3;

```
55 Operation Success!
('201215121', '李勇', '男', 21, 'CS')
('201215122', '刘晨', '女', 20, 'CS')
('201215124', '张立', '男', 20, 'IS')
('201215123', '王敏', '女', 19, 'MA')
56 Operation Success!
(4,)
57 Operation Success!
(0,)
58 Operation Success!
(Decimal('20.0000'),)
59 Operation Success!
(Decimal('27'),)
60 Operation Success!
```

多表查询

SELECT Student.Sno,Sname FROM Student,SC WHERE Student.Sno=SC.Sno AND SC.Cno='2';

```
61 Operation Success!
```

SELECT FIRST.Cno,SECOND.Cpno FROM Course FIRST,Course SECOND WHERE FIRST.Cpno=SECOND.Cno;

SELECT* FROM Student LEFT OUTER JOIN SC ON(Student.Sno=SC.Sno);

SELECT* FROM Course RIGHT OUTER JOIN SC ON(Course.Cno=SC.Cno);

SELECT* FROM Course JOIN SC ON(Course.Cno=SC.Cno);

```
62 Operation Success!
('3', '5')
('6', '5')
('6', '5')
('1', '7')
('5', '6')
('4', '1')
('7', '1')
('4', '1')
('7', '1')
63 Operation Success!
('201215121', '李勇', '男', 21, 'CS', None, None, None)
('201215122', '刘晨', '女', 20, 'CS', None, None, None)
('201215123', '王敏', '女', 19, 'MA', None, None, None)
('201215124', '张立', '男', 20, 'IS', None, None, None)
64 Operation Success!
65 Operation Success!
```

嵌套查询

SELECT Sname FROM Student WHERE EXISTS(SELECT* FROM SC WHERE Sno=Student.Sno AND Cno='1');

```
66 Operation Success!
```

7. 视图

- 定义视图

CREATE VIEW V1 AS SELECT SC.Sno,Sname FROM Student,SC WHERE Sdept='IS' AND SC.Cno='1' AND Student.Sno=SC.Sno;

CREATE VIEW V2 AS SELECT Sname FROM V1;

```
67 Operation Success!
68 Operation Success!
```

- 删除视图

DROP VIEW V2;

```
69 Operation Success!
```

- 查询视图

SELECT Sname FROM V1 WHERE Sname LIKE '李_';

70 Operation Success!

3. XDSQL 数据库优化

（1）源码分析。

其实这次优化的主要代码集中在 tablejoin 这个函数，从 tablejoin 的源码中可以看出，tablejoin 的作用是将 tablenode 当前表和 next 的表首先进行物理数据表信息保存，然后进行二者的连接。连接的过程就是对第一个用户数据表中的每一个元组，都打开第二个表读取所有元组并拼接，将结果写入 temp 临时表。

（2）优化思路。

显然，原本的 tablejoin 只能实现两个表的连接，如果想要实现多个表的连接的话，在不大幅度修改的源码的前提下，我们可以采取递归的策略，将多个表拆成多次两个表连接，即两个表连接的结果返回作为上一次递归连接的对象，两个连接的对象表其中有一个是当前表，另外一个是下一次递归的连接结果返回给当前递归的表对象，这样一层一层层层反复，就能实现。

但是，需要重点的实现的是，在每一次源码实现过程中都需要传递进新的 temp 临时表，所以每一次递归都得产生一个新的 temp 临时表，所以这样就得需要修改 generateTempTable()进行产生新的临时表用来，这样 generateTempTable 函数就需要修改。

（3）代码实现。

```
void tablejoin(tableNode* tablenode, char* row, int index, storageNode* sp,int n) {
	//sp：储存 temp 临时表信息的结构体
	//tablenode：第一个表的表元信息节点



	tableNode* firstnode = tablenode;
	//第一个用户数据表的表名（带有相对路径）
	if (firstnode->next->next != NULL)
	{
		firstnode->next = generateTempTable(firstnode->next, n + 1);
	}
```

```
char* firstfilename = dbnameADDtablename(usedatabase, firstnode->tablename);
storageNode* firstsp = createStorageNode();                    //储存第一个用户数据表信息的结构体
open(firstsp, firstfilename, firstnode->rowlength);
rnd_init(firstsp);


//对第一个用户数据表中的每一个元组，都打开第二个表读取所有元组并拼接，将结果写入 temp 临时表
while (!isEOF(firstsp)) {
    char* first_row = rnd_next(firstsp);                    //读取第一个表的一个元组
    if (first_row == NULL) //读取到被删除数据，返回空值
        continue;

    //将第一张表一行数据放入 row 数组中
    memcpy_s(row, firstnode->rowlength, first_row, firstnode->rowlength);


    //第二个表的表元信息节点
    tableNode* secondnode = firstnode->next;
    char* secondfilename = dbnameADDtablename(usedatabase, secondnode->tablename);
    storageNode* secondsp = createStorageNode();
    open(secondsp, secondfilename, secondnode->rowlength);
    rnd_init(secondsp);
    //遍历第二个表所有的元组，逐个与第一个表中选中的元组拼接
    while (!isEOF(secondsp)) {
        char* second_row = rnd_next(secondsp);
        if (second_row == NULL)
            continue;
        memcpy_s(row + firstnode->rowlength, secondnode->rowlength, second_row, secondnode->rowlength);
        //将第二张表的一行数据添加到 row 数组末尾
        write_row(sp, row);
    }
    close(secondsp);
    deleteStorageNode(secondsp);
```

```
        }

        /* 清除缓存 */
        close(firstsp);
        deleteStorageNode(firstsp);

}

tableNode* generateTempTable(tableNode* tablenode,int n) {
        /* 创建临时表 */
        char tablename[10] = "temp";
        tablename[4] = n + '0';
        char* filename = dbnameADDtablename(usedatabase, tablename);
        if (!create(filename)) {
                printf("create temp error");
                return NULL;
        }

        /* 初始化临时表 tablenode 内列数量和行长度 */
        tableNode* sptablenode = createTableNode();                    /*创建临时表节点*/
        setTableNode(sptablenode, tablename, 0, 0);
        tableNode* node = tablenode;
        while(node != NULL) {                /*计算临时表的列数、行长度*/
                sptablenode->colnum += node->colnum;
                sptablenode->rowlength += node->rowlength;
                node = node->next;
        }

        /* 创建用于记录笛卡尔积中元组的字节数组 row,并打开临时表 */
        char* row = new char[sptablenode->rowlength];
        storageNode* sp = createStorageNode();
        open(sp, filename, sptablenode->rowlength);

        /* 调用笛卡尔积函数 */
        tablejoin(tablenode, row, 0, sp,n);

        /* 清除缓存 */
        close(sp);
        deleteStorageNode(sp);
```

```
        return sptablenode;
}
```

（4）测试结果截图。

```
init SQL ok
SQL>>use database class_db;
use class_db
SQL>>select* from tb1,tb2,tb3,tb4;
tb1.col1        tb1.col2        tb2.col1        tb2.col2        tb3.col1        tb3.col2        tb3.col3        tb4.col1tb4.col2        tb4.col3
1       mike    mike    math    math    1       10      10      20      30
1       mike    mike    math    math    1       10      100     200     300
1       mike    mike    math    physics 2       3       10      20      30
1       mike    mike    math    physics 2       3       100     200     300
1       mike    lee     physics math    1       10      10      20      30
1       mike    lee     physics math    1       10      100     200     300
1       mike    lee     physics physics 2       3       10      20      30
1       mike    lee     physics physics 2       3       100     200     300
1       mike    andi    math    math    1       10      10      20      30
1       mike    andi    math    math    1       10      100     200     300
1       mike    andi    math    physics 2       3       10      20      30
1       mike    andi    math    physics 2       3       100     200     300
2       andi    mike    math    math    1       10      10      20      30
2       andi    mike    math    math    1       10      100     200     300
2       andi    mike    math    physics 2       3       10      20      30
2       andi    mike    math    physics 2       3       100     200     300
2       andi    lee     physics math    1       10      10      20      30
2       andi    lee     physics math    1       10      100     200     300
2       andi    lee     physics physics 2       3       10      20      30
2       andi    lee     physics physics 2       3       100     200     300
2       andi    andi    math    math    1       10      10      20      30
2       andi    andi    math    math    1       10      100     200     300
2       andi    andi    math    physics 2       3       10      20      30
2       andi    andi    math    physics 2       3       100     200     300
3       lee     mike    math    math    1       10      10      20      30
3       lee     mike    math    math    1       10      100     200     300
3       lee     mike    math    physics 2       3       10      20      30
3       lee     mike    math    physics 2       3       100     200     300
3       lee     lee     physics math    1       10      10      20      30
3       lee     lee     physics math    1       10      100     200     300
3       lee     lee     physics physics 2       3       10      20      30
3       lee     lee     physics physics 2       3       100     200     300
3       lee     andi    math    math    1       10      10      20      30
3       lee     andi    math    math    1       10      100     200     300
3       lee     andi    math    physics 2       3       10      20      30
3       lee     andi    math    physics 2       3       100     200     300
Total row: 36
```

4. 遇到的主要问题及解决方法

1. 在实验 1 中，由于没有加“#coding:utf-8”，会出现报错情况，需要添加编码方式才会运行成功。

2. 在实验 2 中，由于是要远程连接阿里云的 PolarDB，所以一条一条输入数据库指令有些麻烦，所以我就想有没有简单的方法来进行，最终我采取的是通过文件的读取和输出来对每个指令的运行，由于有指令存在无法执行的情况，所以还需要用到异常检测，所以我又加上了 try … except …的代码段，不管是异常还是成功都会 有结果输出到文件中。

具体代码可见附录代码

3. 在实验 3 中，要用到递归的算法，需要对源码进行改进优化，这属于牵一发而动全身，所以每一次改变的参数的提取都需要在定义处和引用处都需要进行更改。不然就会出现无法解析，或者参数过多或过少的情况报错。并且代码的编写需要建立在对所用的结构体和某些函数的作用有着充分理解的基础之上，所以需要将源码理解清楚才可进行下一步的编写。

5. 总结

通过这次实验，我有了一定的收获。

第一， 我掌握了远程连接数据库的方法，同时锻炼了我的 python 编程能力。

第二，通过数据库多种操作的实验，我感受到了云数据库和电脑数据库的区别，以及两者的优势所在，从操作的允许和不允许可以明显看出，云数据库更注重数据的安全性，而电脑自己开发的数据库更有用户体验和操作性。两者的功能也有一定的区别，受限于一些权限的问题。同时我用文件来进行指令的输入输出，极大地方便了实验的进行，并且也锻炼了我在这方面的 python 编程能力。

第三，实验 3 对数据库的优化是要建立在对源码的充分理解的基础之上才可以进行，对多定义的结构体含义和某些函数的作用需要有相应的理解，并且其中还夹杂了很多的指针用法，有一定的理解难度和编写难度。同时还需要在不大幅度改变即充分利用已有的代码基础来进行改写，也是一个值得思考的事情，需要对函数每一个引用的对象如何巧妙的转化我改写后的代码流程所需要的参数，这也是一个比较复杂的事情。

附录 实验 2 代码附录和文件截图

```
#coding:utf-8

from typing import ContextManager
import pymysql

connection = pymysql.connect(host='pc-
bp18rn0tqu85a1600-public.rwlb.rds.aliyuncs.com',
                        port=3306,
                        user='lab_1429236414',
                        passwd='515f2c9906a6_#@Aa',
                        db='polardb_mysql_19287cmm')
number = 0
try:
    with connection.cursor() as cursor:
        # 利用文件输入，并且保存输出
         readfile = open('test.txt',"r",encoding='utf-8')
        writefile =
open('result.txt',"w",encoding='utf-8')
        instruction = readfile.readlines()
```

```
        for sql in instruction:
            number += 1
            print(number)
            writefile.write(str(number) + ' ')
            try:
                cursor.execute(sql)
            except Exception as e:
                print(e)
                writefile.write("Something is wrong!
Please look at the terminal immediately!\n")
            else:
                writefile.write("Operation Success!\n")
                for result in cursor:
                    # 输出查询结果
                    # print(result)
                    writefile.write(str(result))
                    writefile.write("\n")

finally:
    writefile.close()
    readfile.close()
    connection.close()
```

test.txt - 记事本

文件(F) 编辑(E) 格式(O) 查看(V) 帮助(H)

```
CREATE TABLE Course(Cno CHAR(4),Cname CHAR(40),Cpno CHAR(4) , Ccredit INT);
CREATE TABLE SC(Sno CHAR(9),Cno CHAR(4),Grade INT);
INSERT INTO Student VALUES('201215121','李勇','男',20,'CS');
INSERT INTO Student VALUES('201215122','刘晨','女',19,'CS');
INSERT INTO Student VALUES('201215123','王敏','女',18,'MA');
INSERT INTO Student VALUES('201215124','张立','男',19,'IS');
INSERT INTO COURSE VALUES('1','数据库','5',4);
INSERT INTO COURSE VALUES('3','信息系统','1',4);
INSERT INTO COURSE VALUES('4','操作系统','6',3);
INSERT INTO COURSE VALUES('5','数据结构','7',4);
INSERT INTO COURSE VALUES('7','PASCAL语言','6',4);
INSERT INTO SC VALUES('201215121','1',92);
INSERT INTO SC VALUES('201215121','2',85);
INSERT INTO SC VALUES('201215121','3',88);
INSERT INTO SC VALUES('201215122','2',90);
INSERT INTO SC VALUES('201215122','3',80);
INSERT INTO COURSE VALUES('2','数学',NULL,2);
INSERT INTO COURSE VALUES('6','数据处理',NULL,2);
insert into course(Cno,Cname,Ccredit) values ('2','数学',2);
insert into course(Cno,Cname,Ccredit) values ('6','数据处理',2);
UPDATE Student SET Sage=Sage+1;
UPDATE Course SET Cpno='1' WHERE Cno='6';
DELETE FROM SC WHERE Sno='201215121' AND Cno='1';
DELETE FROM SC;
SELECT * FROM Student;
SELECT * FROM Sc;
SELECT * FROM Course;
SELECT Sno,Sname,Sage FROM Student;
SELECT Sname FROM Student WHERE Sdept='cs';
SELECT Sname,Sage FROM Student WHERE Sage<20;
SELECT Sname,2021-Sage FROM Student;
SELECT Sname,'Year of birth:',2014-Sage FROM Student;
SELECT DISTINCT Sno FROM SC;
SELECT ALL Sno FROM SC;
SELECT* FROM Student WHERE Sage BETWEEN 20 AND 21;
SELECT * FROM Student WHERE Sdept IN('CS','IS');
SELECT * FROM Student WHERE Sname LIKE '刘%';
SELECT * FROM Student WHERE Sname LIKE '刘_';
SELECT* FROM Student ORDER BY Sdept,Sage DESC;
SELECT COUNT(*) FROM Student;
SELECT COUNT(DISTINCT Sno) FROM SC;
SELECT AVG(Sage) FROM Student;
SELECT SUM(Ccredit) FROM Course;
SELECT Sno FROM SC GROUP BY SNO HAVING COUNT(*)>=3;
SELECT Student.Sno,Sname FROM Student,SC WHERE Student.Sno=SC.Sno AND SC.Cno='2';
SELECT FIRST.Cno,SECOND.Cpno FROM Course FIRST,Course SECOND WHERE FIRST.Cpno=SECON
SELECT* FROM Student LEFT OUTER JOIN SC ON(Student.Sno=SC.Sno);
SELECT* FROM Course RIGHT OUTER JOIN SC ON(Course.Cno=SC.Cno);
SELECT* FROM Course JOIN SC ON(Course.Cno=SC.Cno);
SELECT Sname FROM Student WHERE EXISTS(SELECT* FROM SC WHERE Sno=Student.Sno AND Cı
CREATE VIEW V1 AS SELECT SC.Sno,Sname FROM Student,SC WHERE Sdept='IS' AND SC.Cno='1'
CREATE VIEW V2 AS SELECT Sname FROM V1;
DROP VIEW V2;
SELECT Sname FROM V1 WHERE Sname LIKE '李_';
```

第 1 行，第 1 列 100% Windows (CRLF) UTF-8

result.txt - 记事本

文件(F) 编辑(E) 格式(O) 查看(V) 帮助(H)

```
1 Something is wrong! Please look at the terminal immediately!
2 Something is wrong! Please look at the terminal immediately!
3 Operation Success!
4 Operation Success!
5 Operation Success!
6 Something is wrong! Please look at the terminal immediately!
7 Something is wrong! Please look at the terminal immediately!
8 Something is wrong! Please look at the terminal immediately!
9 Something is wrong! Please look at the terminal immediately!
10 Something is wrong! Please look at the terminal immediately!
11 Something is wrong! Please look at the terminal immediately!
12 Something is wrong! Please look at the terminal immediately!
13 Something is wrong! Please look at the terminal immediately!
14 Something is wrong! Please look at the terminal immediately!
15 Something is wrong! Please look at the terminal immediately!
16 Something is wrong! Please look at the terminal immediately!
17 Something is wrong! Please look at the terminal immediately!
18 Something is wrong! Please look at the terminal immediately!
19 Operation Success!
20 Operation Success!
21 Operation Success!
22 Operation Success!
23 Operation Success!
24 Operation Success!
25 Operation Success!
26 Operation Success!
27 Operation Success!
28 Operation Success!
29 Operation Success!
30 Operation Success!
31 Operation Success!
32 Operation Success!
33 Operation Success!
34 Operation Success!
35 Operation Success!
36 Operation Success!
37 Operation Success!
38 Operation Success!
39 Operation Success!
40 Operation Success!
41 Operation Success!
('201215121', '李勇', '男', 21, 'CS')
('201215122', '刘晨', '女', 20, 'CS')
('201215123', '王敏', '女', 19, 'MA')
('201215124', '张立', '男', 20, 'IS')
42 Operation Success!
43 Operation Success!
('1', '数据库', '5', 4)
('3', '信息系统', '1', 4)
('4', '操作系统', '6', 3)
('5', '数据结构', '7', 4)
('7', 'PASCAL语言', '6', 4)
('2', '数学', None, 2)
('6', '数据处理', '1', 2)
('2', '数学', None, 2)
('6', '数据处理', '1', 2)
44 Operation Success!
('201215121', '李勇', 21)
('201215122', '刘晨', 20)
('201215123', '王敏', 19)
('201215124', '张立', 20)
45 Operation Success!
('李勇',)
('刘晨',)
46 Operation Success!
('王敏', 19)
47 Operation Success!
('李勇', 2000)
('刘晨', 2001)
('王敏', 2002)
('张立', 2001)
48 Operation Success!
('李勇', 'Year of birth:', 1993)
('刘晨', 'Year of birth:', 1994)
('王敏', 'Year of birth:', 1995)
('张立', 'Year of birth:', 1994)
49 Operation Success!
50 Operation Success!
51 Operation Success!
('201215121', '李勇', '男', 21, 'CS')
('201215122', '刘晨', '女', 20, 'CS')
('201215124', '张立', '男', 20, 'IS')
52 Operation Success!
('201215121', '李勇', '男', 21, 'CS')
('201215122', '刘晨', '女', 20, 'CS')
('201215124', '张立', '男', 20, 'IS')
53 Operation Success!
('201215122', '刘晨', '女', 20, 'CS')
54 Operation Success!
('201215122', '刘晨', '女', 20, 'CS')
55 Operation Success!
('201215121', '李勇', '男', 21, 'CS')
('201215122', '刘晨', '女', 20, 'CS')
('201215124', '张立', '男', 20, 'IS')
('201215123', '王敏', '女', 19, 'MA')
56 Operation Success!
(4,)
57 Operation Success!
(0,)
58 Operation Success!
(Decimal('20.0000'),)
59 Operation Success!
(Decimal('27'),)
60 Operation Success!
61 Operation Success!
62 Operation Success!
('3', '5')
('6', '5')
('6', '5')
('1', '7')
('5', '6')
('4', '1')
('7', '1')
('4', '1')
('7', '1')
63 Operation Success!
('201215121', '李勇', '男', 21, 'CS', None, None, None)
('201215122', '刘晨', '女', 20, 'CS', None, None, None)
('201215123', '王敏', '女', 19, 'MA', None, None, None)
('201215124', '张立', '男', 20, 'IS', None, None, None)
64 Operation Success!
65 Operation Success!
66 Operation Success!
67 Operation Success!
68 Operation Success!
69 Operation Success!
70 Operation Success!
```